

Radio Telescope Project

2025-10-09 Meeting Notes

Introductions

Present: Dr. Wenger, Lily, Alex, Anthony

Why build a radio telescope?

- Because it's fun!
- Interdisciplinary collaboration.
- Useful education and public outreach tool.
- Low risk, high-reward.

How to build a radio telescope?

Here are some links to radio telescope kits, parts lists, and construction guides.

- The original Small Radio Telescope (SRT) from MIT, which has been discontinued:
<https://www.haystack.mit.edu/haystack-public-outreach/srt-the-small-radio-telescope-for-education/#current>
- SRT suggests building this “TV Dongle” version now:
https://www.haystack.mit.edu/wp-content/uploads/2020/07/memo_SRT_025.pdf
- As a cheaper alternative, we could just build a “horn” telescope:
https://www.haystack.mit.edu/wp-content/uploads/2020/07/memo_SRT_027.pdf
- Another “horn” design from SKAO:
<https://www.astronomy.me.uk/wp-content/uploads/2025/02/SKAO-Table-Top-RT-How-to-Guide-acc-100225.pdf>
- Another “horn” design from DSPIRA: <https://wvurail.org/dspira-lessons/>

Define steps and assign roles

We brainstormed the following steps/components of the telescope. These are the different “project groups”:

1. **The dish/horn:** we need to secure or build a dish or horn. Ideas include: looking for old satellite dishes that people are giving away, look for cheap options on eBay or elsewhere. A backup plan is to build a horn.
2. **The mount:** ideally we would have an alt/az telescope mount that is capable of moving the >100 pound (maybe up to 300 pounds?) dish. It does not have to be precise since radio waves are big. The telescope could be placed on a fixed mount until a tracking

mount is built/available. There are some “off the shelf” tracking mount options, but they are expensive.

3. **The receiver electronics:** the above guides describe the necessary components (software defined radio, low-noise amplifiers, etc.)
4. **The control electronics:** we’ll also need to control the mount, calibration source, etc.
5. **The software:** the above guides describe the available software, but we will probably have to write some of our own.

Here are the roles we identified:

1. **Project leader:** coordinate between different project groups
2. **Faculty advisor:** Dr. Wenger will help interface with the University and manage bureaucracy
3. **Fundraising coordinator:** someone to manage fundraising
4. **Group leaders:** people to take charge of the project groups
5. **Group members:** people to help in each group

Some clubs that we should reach out to for volunteers:

1. Physics
2. Astronomy
3. Rocketry
4. SME
5. Aime

Other Business

- Need to think of a cool project name
- Fundraising opportunity: giving day
- Need to get more volunteers
- Sign up for roles here:
<https://docs.google.com/spreadsheets/d/1SwEwVaqYb8b5dP4kuRnIhATOi4gfXdtsg76lpVHdKbU/edit?usp=sharing>